

OceanSense: An IoT-Based Hybrid System for Real-Time Detection and Tracking of Marine Debris

25-26J-002

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Computer Systems &
Network Engineering

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

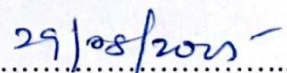
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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

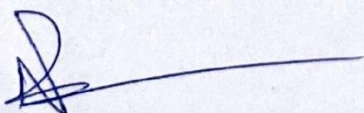


Signature of the Supervisor

(Ms. Dinithi Pandithage)

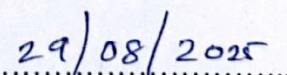


Date



Signature of the Co-Supervisor

(Ms. Narmada Gamage)



Date

Abstract

Marine debris monitoring benefits from swarms of surface vehicles that can cover large areas and operate continuously[1]. However, such swarms face challenges from dynamic ocean conditions (waves, currents) and severely limited communication bandwidth. We propose a decentralized swarm coordination model for ASVs that adapts formation and movement based on local environmental states (e.g. wave height, current direction, debris density). Communication is handled exclusively via low-power LoRa radios, which provide long-range links at low data rates[2][3]. The novelty lies in using real-time environment cues to adjust swarm behavior under LoRa constraints, enhancing coverage and robustness. We will implement the model in simulation (Webots + ROS) and evaluate it on metrics such as area coverage efficiency, formation stability under changing conditions, LoRa message delivery rate, node-failure recovery time, and energy usage. This adaptive formation approach aims to improve decentralized coordination and resilience in marine debris tracking tasks.

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List of Abbreviations

AUV: Autonomous Underwater Vehicle
ASV: Autonomous Surface Vehicle
DC: Decentralized Coordination
IoT: Internet of Things
LPWAN: Low-Power Wide-Area Network
LoRa: Long Range (radio communication technology)
LoRaWAN: LoRa Wide Area Network
ML: Machine Learning
MIMO: Multiple-Input Multiple-Output
MPC: Model Predictive Control
PID: Proportional–Integral–Derivative (controller)
RL: Reinforcement Learning
ROS: Robot Operating System
RSSI: Received Signal Strength Indicator
RT: Recovery Time
SI: Swarm Intelligence
SLAM: Simultaneous Localization and Mapping

1. Introduction

1.1 Background & Literature Survey

Marine debris tracking and water quality monitoring increasingly rely on autonomous vehicles. Unmanned Surface Vehicles (USVs) and other UV platforms (UAVs, gliders) have shown great promise in covering diverse aquatic areas for environmental sensing[1]. These systems offer flexibility and broad spatial coverage compared to fixed sensing or satellite methods. Concurrently, the Internet of Things (IoT) enables embedding sensors in debris or vehicles; for example, a “smart bottle” with a LoRa IoT sensor was able to track plastic waste pathways without GPS[4]. LoRa radios, in particular, have emerged as a key technology for long-range, low-power communication between IoT nodes[2]. Recent work by Souli *et al.* demonstrated a ROS-LoRa UAV swarm for disaster response, leveraging LoRa’s long-range links to share data between drones[3].

Swarm robotics is well-suited for area coverage tasks in uncertain environments. Decentralized swarms exhibit high scalability and robustness, handling node failures gracefully[5]. However, most swarm control strategies assume either high-bandwidth links or ignore rich environmental feedback. Traditional swarm models (e.g. flocking, potential fields) typically use fixed behaviors that do not explicitly adapt to measured wave or current conditions. Furthermore, LoRa’s low data rate (hundreds of bits/s) and duty-cycle limits mean only sparse messages can be exchanged[2]. Thus, there is a need for novel algorithms that combine bio-inspired swarm coordination with adaptation to local marine states while operating under strict communication constraints.

1.2 Research Gap

Prior studies on robotic coverage have explored many algorithms for maximizing area coverage and foraging, but few incorporate environmental-state cues into the control laws. Additionally, most swarm frameworks assume reliable communication; in contrast, LoRa imposes severe bandwidth and timing limits[2]. The literature lacks a methodology for explicitly using wave height, current direction, and debris density in the swarm’s decision-making. Current approaches typically use fixed formations or reactive obstacle avoidance, rather than proactively adjusting the formation geometry based on ocean conditions. This leads to suboptimal coverage and reduced efficiency when conditions change. Our research gap is the absence of an *environment-based adaptive formation control* scheme for ASV swarms that works with ultra-low-rate LoRa messaging. We address how to make each ASV use its local state measurements to influence the group behavior in a decentralized way, without relying on constant updates from neighbors.

1.3 Research Problem

The core research problem is: **How can a swarm of ASVs coordinate in a decentralized manner to efficiently track marine debris, using only local**

environmental measurements? Specifically, we must design a distributed control strategy that takes as input local environment-state variables (e.g. significant wave height, current vector, detected debris density) and computes motion commands that both cover the area and maintain a stable formation. The swarm should maintain high coverage of the search region while responding to changes (e.g. re-spacing in high waves or re-orienting along currents) and coping with node losses. All coordination must occur via LoRa messages (broadcast or mesh) with limited payloads. We will create a simulation framework to test whether such an adaptive strategy can outperform non-adaptive baselines in coverage metrics and resilience under realistic maritime conditions.

2. Objectives

2.1 Main Objective

The main objective of this project is to **develop and validate a decentralized, environment-state-based adaptive swarm coordination model for a fleet of ASVs**, specifically tailored for marine debris tracking, using only LoRa communication. The model will allow ASVs to adjust their formation and search patterns in response to dynamic ocean conditions while still achieving effective area coverage and resilience to individual node failures.

2.2 Sub-objectives

- Develop a mathematical model for ASV swarm behavior that incorporates local environment-state measurements (wave height, current, debris density) into formation and motion control.
- Design decentralized communication protocols over LoRa that allow minimal information (e.g. event flags or compressed data) to be exchanged for coordination.
- Implement the swarm control algorithm and communications in a simulation environment using Webots and ROS, including models of waves, currents, and sparse LoRa links.
- Define use-case scenarios (e.g. high-wave environment, uneven debris distribution) and evaluation metrics (area coverage, formation stability, LoRa success rate, recovery time).
- Run experiments in simulation to tune parameters and verify that the adaptive swarm outperforms static strategies, achieving higher coverage and faster recovery after node failures.
- Document the system design, simulation results, and analysis, culminating in a complete project report and potential conference/journal submission.

3. Methodology

This project will be executed in four phases, combining design, development, integration, and documentation. Core tools include the open-source Webots robotics simulator and ROS (Robot Operating System) for control logic.

3.1 Phase 1: Design & Planning

- **Requirement Analysis:** Define system scope and identify sensor inputs (wave height sensor, current vector sensor, debris detector) and expected outputs (velocity and heading commands, formation adjustments).
- **Environment Modeling:** Establish a simulation of ocean conditions (e.g. sinusoidal wave fields, uniform currents, random debris fields) in Webots. Determine realistic ranges for wave height and current that ASVs will measure.
- **Swarm Architecture:** Decide on swarm control approach (e.g. virtual structure or leader-follower) and how to parameterize formation (spacing, shape) based on environment-state. Draft algorithms for formation adaptation (e.g. increase inter-vehicle spacing when wave height is high).
- **Communications Planning:** Design LoRa-based messaging schemes for the swarm (e.g. periodic status broadcasts, event triggers). Establish message contents (e.g. position, local state summaries) subject to LoRa payload limits.
- **Evaluation Criteria:** Define performance metrics: area coverage percentage over time, average spacing error (formation stability), message delivery success rate, recovery time after a simulated node failure, and energy consumption (propulsion + comm).
- **Project Schedule:** Create a detailed timeline for development tasks, setting milestones for each phase.

3.2 Phase 2: Development & Implementation

- **Simulator Setup:** Configure Webots simulation world with surface vehicles and dynamic water models. Integrate ROS nodes controlling each ASV through the Webots ROS bridge.
- **Algorithm Coding:** Implement the decentralized control algorithms in ROS. Each ASV node reads simulated sensor data (e.g. wave height, local debris density) and adjusts its target waypoint or formation offset accordingly. E.g.: if debris density high nearby, slow down or form cluster; if wave height exceeds threshold, increase distance from neighbors to reduce collisions.
- **LoRa Communication Emulation:** Create a communication layer in ROS that simulates LoRa constraints: low-bandwidth broadcasts at intervals, possible packet drops, and duty-cycle enforcement. Use ROS topics or services to represent LoRa messages among ASV nodes.
- **Integration:** Combine control and communication: ensure that each ASV can make decisions with only periodic information from others (e.g. neighbor positions) and mostly relies on own sensors. Validate that the algorithm runs without a central coordinator.

- **Initial Testing:** Run simple scenarios (e.g. calm water, no current, uniform debris) to debug the basic swarm behavior and ensure vehicles can form the intended pattern and maintain connectivity.

3.3 Phase 3: Integration & Evaluation

- **Scenario Testing:** Execute simulated missions in more challenging conditions (varying wave heights, directional currents, clustered debris). For each scenario, collect data on the defined metrics.
- **Performance Analysis:** Measure **area coverage efficiency** (percentage of region surveyed within time), **formation stability** (variance of inter-ASV distances), **LoRa message success rate** (ratio of successful transmissions), **recovery time** (time to reform coverage after one ASV is disabled), and **energy use** (estimated from velocity and communication duty cycle).
- **Baseline Comparison:** Compare the adaptive strategy against a non-adaptive baseline (e.g. fixed formation or random walk) to quantify improvements.
- **Robustness Checks:** Simulate ASV failures or communication blackout events to test resilience. For example, disable one ASV mid-mission and observe how quickly neighbors fill the gap.
- **Parameter Tuning:** Adjust control gains, communication rates, and adaptation thresholds to optimize performance. Use iterative testing to refine algorithms.
- **Documentation of Results:** Record experiment setups and outcomes. Generate plots of coverage over time, formation snapshots, and recovery dynamics.

3.4 Phase 4: Documentation & Dissemination

- **Reporting:** Compile a comprehensive project report covering all phases, methodology, and results. Include figures and tables illustrating key findings.
- **Code Documentation:** Comment and clean the ROS/Webots codebase. Prepare user documentation so others can reproduce the simulations.
- **Proposal & Presentation:** Finalize this research proposal draft in SLIIT format. Prepare any required presentations or poster for the project defense.
- **Publications (optional):** If results are strong, draft a conference or journal paper focusing on the novel swarm algorithm and its performance under LoRa constraints.

4. Project Requirements

4.1 Functional Requirements

- **Environmental Sensing:** Each ASV must measure local wave height, current direction/magnitude, and debris density (e.g. via onboard sensors or vision).
- **Area Coverage:** The swarm must systematically cover a designated marine area to locate debris.
- **Adaptive Formation Control:** The swarm's formation geometry (spacing and orientation) must adjust in real time based on sensed environmental conditions.
- **Decentralized Coordination:** No central controller; each ASV independently computes its control actions using local data and intermittent LoRa messages from neighbors.
- **LoRa Communication:** ASVs communicate via LoRa radios. They broadcast position and key status updates at low frequency to coordinate.
- **Resilience:** The system must detect a neighboring ASV's failure (e.g. missing heartbeat) and reassign coverage responsibilities autonomously.
- **Data Logging:** The simulation should record mission data (e.g. positions, sensor readings) for post-analysis.

4.2 Non-Functional Requirements

- **Low Bandwidth Usage:** Communication over LoRa must use minimal payloads; algorithms should function even with dropped or delayed messages.
- **Scalability:** The approach should extend to swarms of tens of ASVs without centralized bottlenecks.
- **Real-time Operation:** Control loops must run in (simulated) real time; computation per ASV should be light enough for low-power processors.
- **Energy Efficiency:** ASV motion and communication strategies should minimize power draw (e.g. using slow speeds in high waves to conserve energy).
- **Reliability:** The system should maintain operation despite individual node failures or harsh conditions.
- **Open-Source Implementation:** Use free tools (Webots, ROS) for cost-effectiveness and transparency.

4.3 Use Cases

- **Use Case 1 – Area Survey:** Deploy the swarm to survey a region for debris. Each ASV follows its assigned segment; if local debris detection is high, it signals neighbors to converge. The swarm adapts its pattern to ensure full coverage while avoiding collisions.
- **Use Case 2 – Rough Seas Adaptation:** In a high-wave scenario, the swarm recognizes increased wave heights via sensors. ASVs autonomously increase spacing and reduce speed to maintain stability. They still communicate occasionally via LoRa to keep track of formation integrity under these conditions.

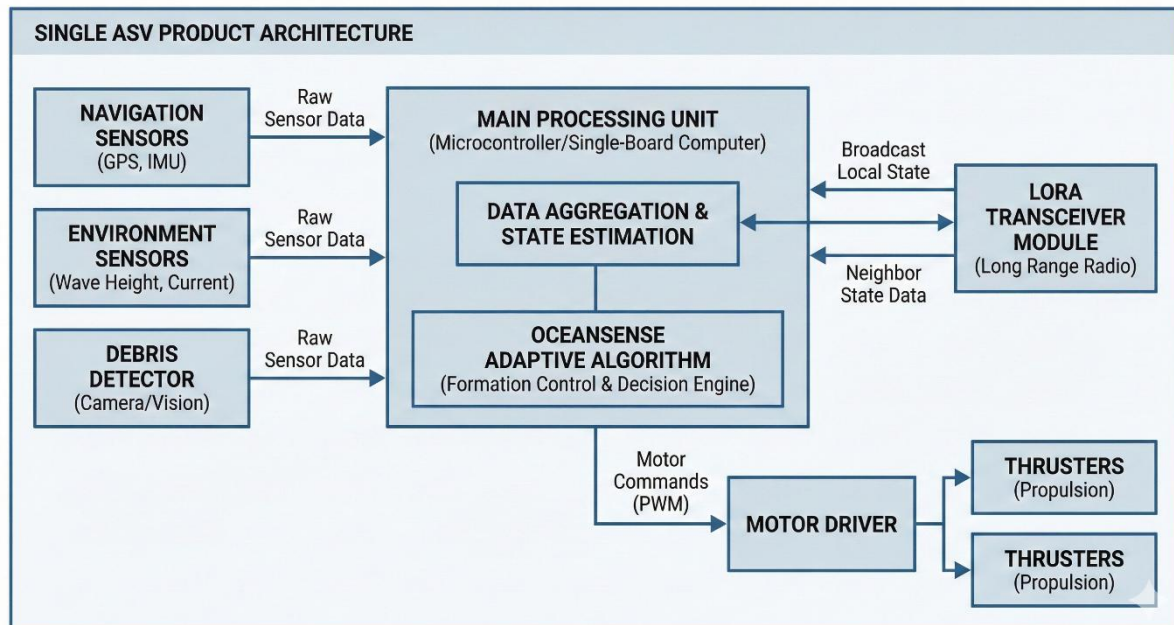
- **Use Case 3 – Current Alignment:** The environment imposes a steady current. The swarm detects current vectors and re-orient its formation (e.g. aligning into the current) to optimize energy usage and search efficiency.
- **Use Case 4 – Node Failure Recovery:** If one ASV runs out of power or suffers malfunction, it stops broadcasting its heartbeat. Neighboring ASVs detect its absence (no LoRa messages) and redistribute its search area among themselves, reforming the formation to cover the gap without human intervention.
- **Use Case 5 – Sparse Communication:** In a scenario with intermittent LoRa connectivity (e.g. due to duty cycle limits), ASVs operate under delayed updates. They rely heavily on local sensing and only use LoRa for crucial alerts (e.g. debris cluster found) to minimize collisions and maximize coverage despite communication sparsity.

5. Budget and Justification

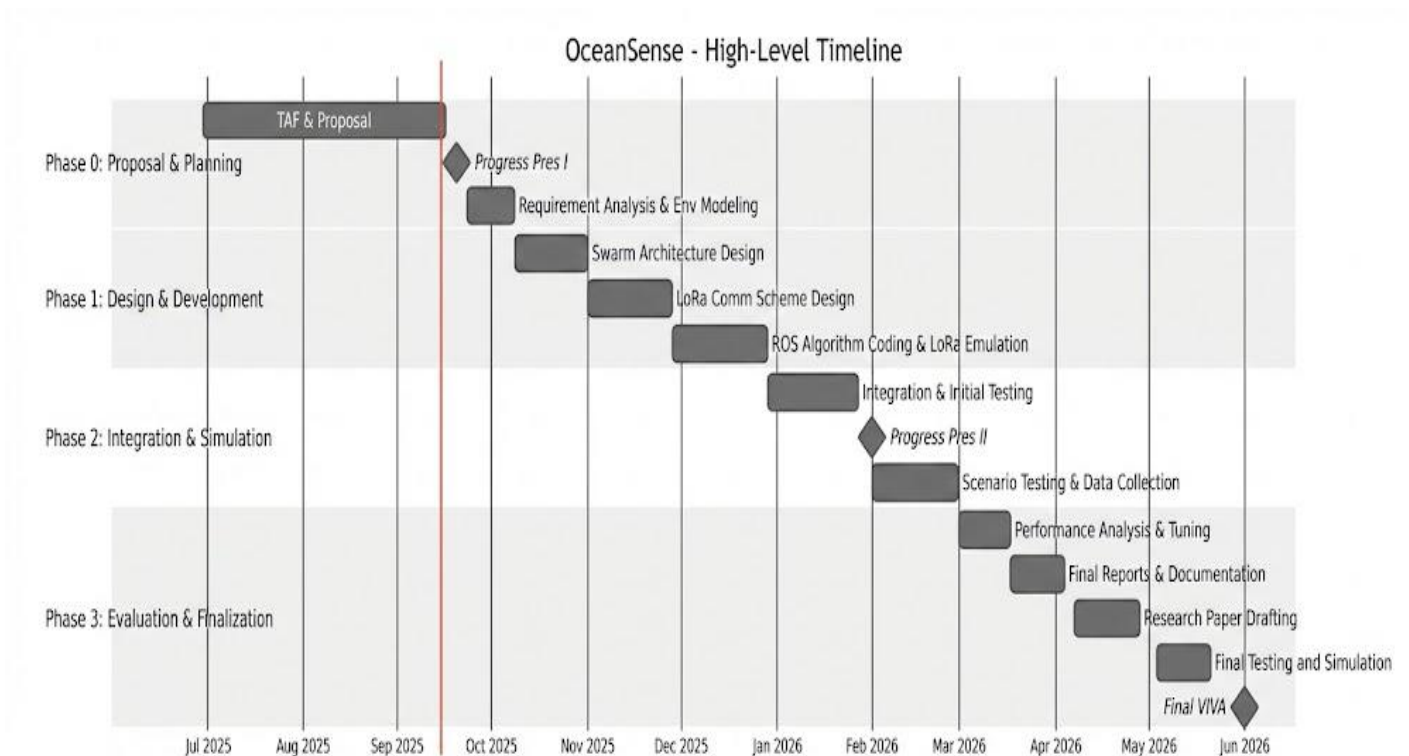
- **Software (free):** Webots simulator (open-source), ROS middleware, Python/C++ libraries. These tools are open-source and have no licensing costs.
- **Computing Resources:** Existing PCs or laptops for simulation; no additional purchase needed.
- **Hardware (minimal):** If a hardware testbed is built, we can reuse departmental ASV platforms or buy a few inexpensive components: e.g. LoRa transceiver modules (3000LKR each) and microcontrollers (15000 each). Any sensors (GPS, IMU) are already standard on test vehicles. We assume up to 3 nodes for proof-of-concept, so total ≈ 25000 per each node in new hardware.
- **Total Estimated Cost:** On the order of a few thousands at most (mostly from potential hardware), which is minimal for a research project of this scope.

6.Appendices

Appendix A : High level architecture diagram



Appendix B : Gantt Chart



7. References

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